XAVIER INSTITUTE OF ENGINEERING
(An Autonomous Institute under Mumbai University)
Approved by AICTE and recognized by DTE, Government of Maharashtra
Accredited with an "A+" grade by NAAC
Department of Computer Science and Engineering
(Internet of Things and Cyber Security including Block Chain Technology)

Mini Project - 2B

Domain: Blockchain and Securtiy Model

Group Members:-

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Under the Guidance of : Dr. Ninad More

Topic 1: Blockchain based Voting System

Problem statement:

Traditional voting systems lack transparency and are vulnerable to fraud, manipulation, and centralized control. Ensuring vote security, integrity, and voter trust remains a major challenge. A blockchain-based voting system provides a decentralized and tamper-proof solution to address these issues.

Introduction:

Voting is a fundamental process in a democratic system, allowing citizens to participate in governance and decision-making. Traditional voting methods, including paper ballots and electronic voting machines, face challenges such as lack of transparency, centralized control, high costs, and vulnerability to fraud or tampering. These issues can reduce public trust in the electoral process.

Blockchain technology provides an effective solution to these challenges by offering a decentralized, secure, and immutable platform for voting. In a blockchain-based voting system, each vote is recorded as a transaction on a distributed ledger and secured using cryptographic techniques. Smart contracts automate vote casting and counting while maintaining voter anonymity and transparency. This system enhances election security, minimizes manipulation, and enables faster and verifiable results.

Abstract:

A blockchain-based voting system uses decentralized ledger technology to provide a secure, transparent, and tamper-proof voting process. Each vote is recorded as an immutable transaction and verified using cryptographic methods, ensuring data integrity and preventing manipulation. Smart contracts automate vote casting and counting while maintaining voter anonymity. This system reduces fraud, increases trust, and enables faster and verifiable election results compared to traditional voting methods.

Software and Algorithms used:

Software:
Blockchain: Ethereum
Language: Solidity
Tools: Remix IDE, Ganache
Frontend: HTML, CSS, JavaScript
Backend: Node.js
Wallet: MetaMask

Algorithms:
Consensus: Proof of Stake / Proof of Work
Hashing: SHA-256
Security: ECDSA (Digital Signature)
Validation: One-Vote-Per-Voter Logic

Thank you